
PRESCRIBING TRENDS ANALYSIS

(ANA-3)

Antidepressant and Anxiolytic Prescribing Trends from OpenPrescribing Data by NHS Organisations

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Project: AI Mental Health Assistant

NHS PRESCRIBING TRENDS ANALYSIS

1. OBJECTIVE

To build a safe and highly responsive triage engine for our AI Mental Health Assistant, we need to understand the real-world clinical baseline of our users. While ANA-1 and ANA-2 analyzed physical therapy wait times and drop-off rates, ANA-3 focuses on pharmaceutical dependency.

By mapping the 5-year prescribing trends of antidepressants and anxiolytics across regional Sub-ICBs, we can identify "clinical hotspots." This allows our AI to adapt its conversational logic—such as prioritizing medication side-effect check-ins or pushing digital Cognitive Behavioral Therapy (dCBT) modules to users in regions heavily reliant on pharmaceuticals.

2. DATASETS OVERVIEW:

Primary Dataset:

OpenPrescribing Mental Health Medication Data

Data Source:

OpenPrescribing — University of Oxford / NHS prescribing data

Dataset Type:

Structured prescribing transaction dataset

Key Medication Categories:

- Antidepressants
- Anxiolytics

Dataset Characteristics:

Attribute	Observation
Data Type	Monthly prescribing data
Coverage	NHS prescribing organisations
Measures Included	Prescription items and prescribing cost
Time Component	Monthly longitudinal records
Data Format	CSV / API JSON
Granularity	Organisation-level prescribing

3. METHODOLOGY:

To bypass potential API connectivity issues with the OpenPrescribing live servers, the data was programmatically extracted and merged using Python from localized CSV datasets. The pipeline isolated two distinct BNF therapeutic classes:

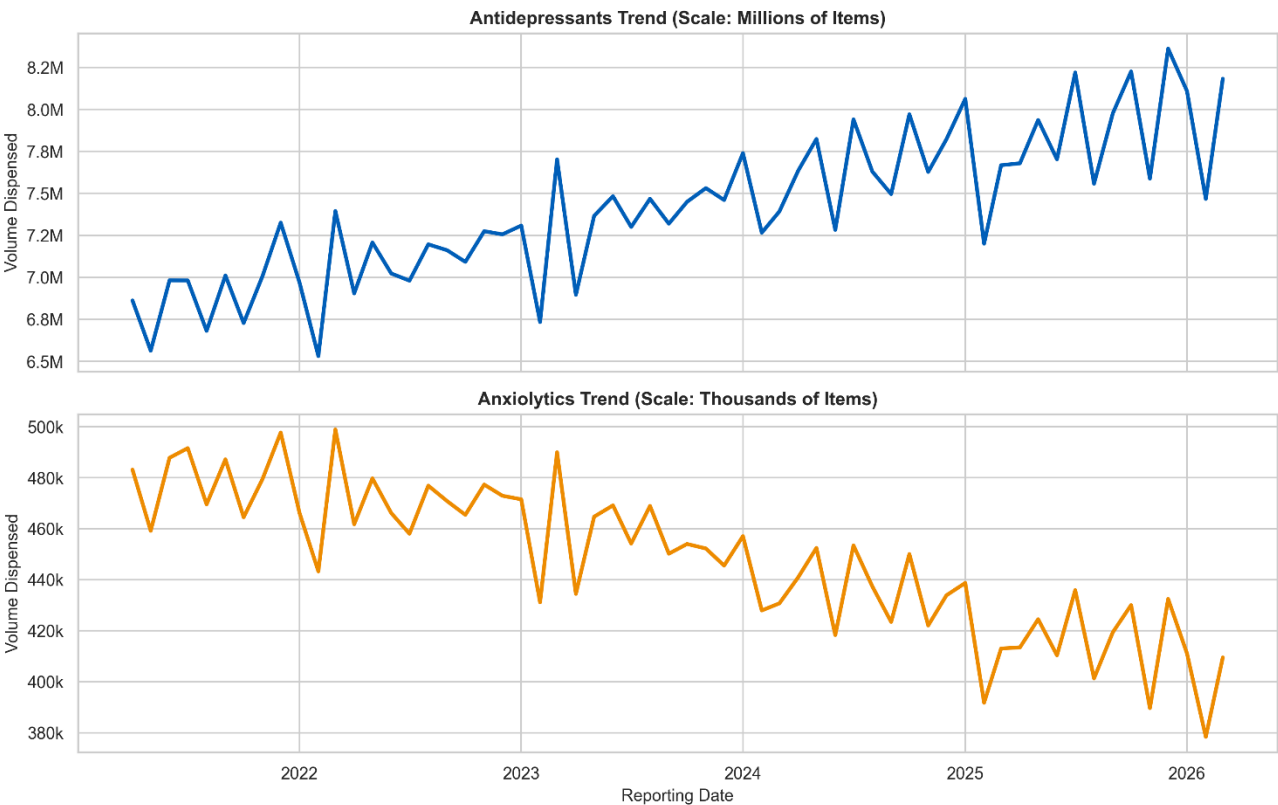
- **Antidepressants:** SSRIs, SNRIs, and Tricyclics.
- **Anxiolytics:** Anti-anxiety medications (excluding hypnotics/sleep aids to prevent data skew).

The datasets were merged, cleaned of zero-value anomalies, and mapped to geographic CCGs to perform time-series, volume, and cost-correlation analyses.

4. KEY FINDINGS: PRESCRIBING TRENDS:

- **The Surging Baseline:** Antidepressant usage is experiencing a sustained, linear increase. Over our 5-year window, national monthly dispensing climbed from 6.86 million items to 8.18 million items. Overall medication volume grew by 16.97%.
- **The Volume Discrepancy:** Are anxiolytics prescribed less frequently than antidepressants? Yes, drastically. The total 5-year volume for antidepressants was a massive 444.7 million items, compared to just 26.8 million items for anxiolytics. This indicates that long-term mood management (antidepressants) is the dominant pharmacological strategy in the UK, rather than acute anxiety management.

Monthly Prescribing Trends: Antidepressants vs Anxiolytics (2021-2026)

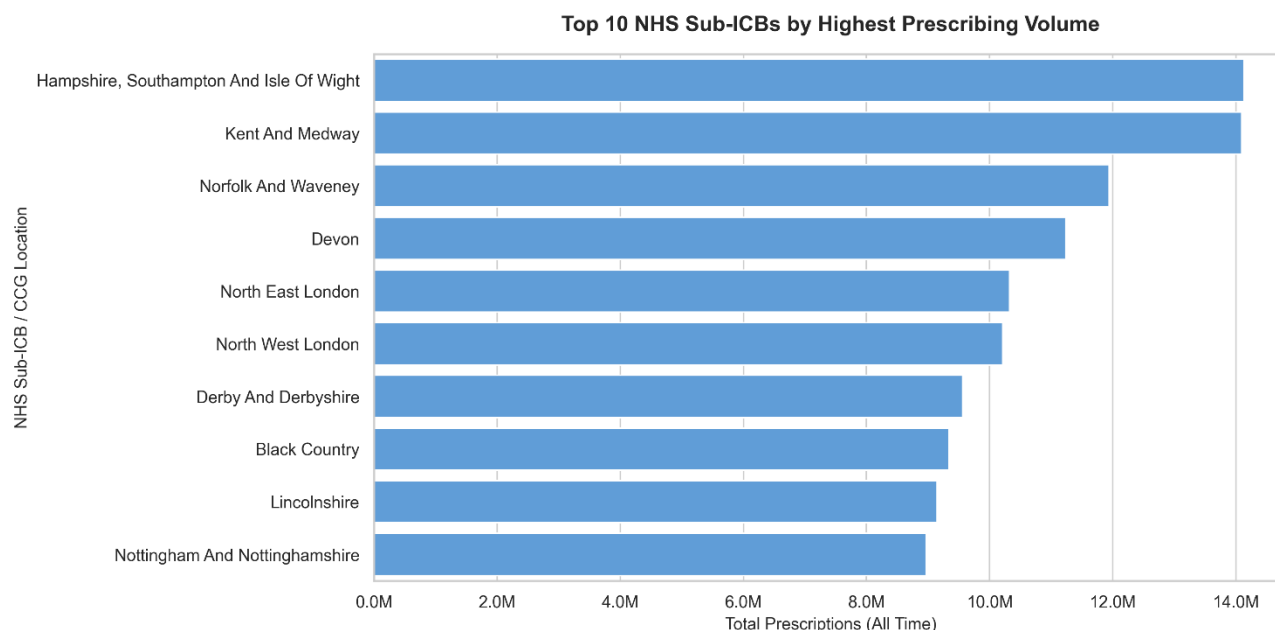


5. KEY FINDINGS: REGIONAL HOTSPOTS & INEQUALITIES:

Prescribing is geographically uneven, highlighting structural inequalities in community mental health management.

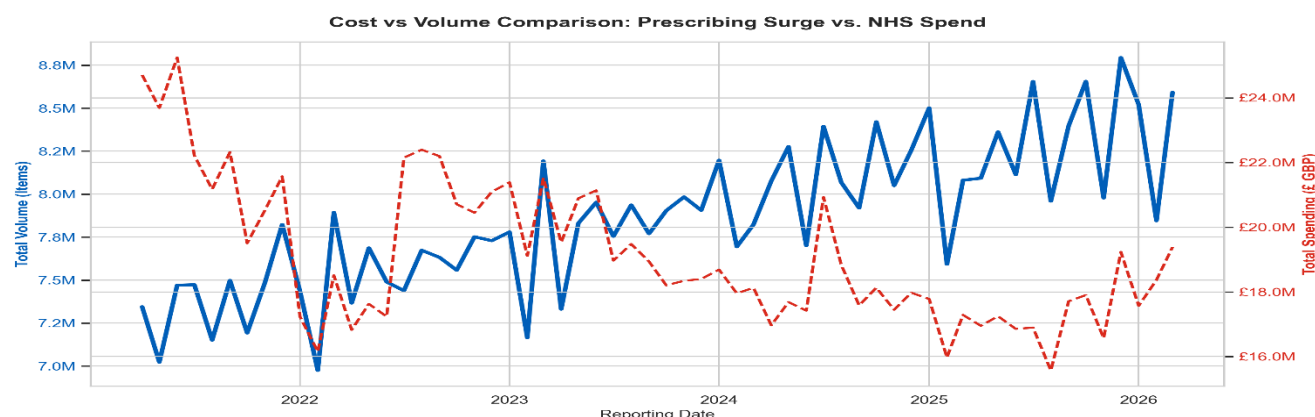
- **Highest Absolute Volume:** Large commuter belts and coastal regions carry the heaviest clinical burdens. The top prescribing Sub-ICBs are Hampshire, Southampton & Isle of Wight (13.4M items) and Kent & Medway (13.3M items), followed closely by Norfolk & Waveney and Devon.

- **Anxiolytic Concentrations:** Interestingly, the regions with the highest anxiolytic usage perfectly mirror the top antidepressant regions (Devon, Norfolk, Kent). This suggests a generalized systemic reliance on medication in these specific geographies.
- **Fastest Surging Regions:** While coastal areas hold the highest *total* volume, the fastest *growth* is happening in urban centers. North Central London saw a staggering 38.8% increase in prescribing volume over 5 years, followed by Thames Valley at 33.5%.



6. KEY FINDINGS: COST DYNAMICS:

- **Total Spend:** Over the 5-year window, the NHS spent £1.06 Billion on antidepressants and £88.5 Million on anxiolytics.
- **Does higher volume mean higher costs?** Surprisingly, no. The correlation matrix between volume and cost returned a value of -0.358 (a negative correlation). *Chart 3* clearly demonstrates that while prescription counts steadily rise, actual NHS spending frequently drops or flatlines. This is due to bulk NHS procurement efficiencies and the widespread use of off-patent, highly affordable generic medications (like Sertraline and Citalopram).



7. CONCLUSION:

The ANA-3 prescribing trends analysis successfully explored antidepressant and anxiolytic prescribing activity using OpenPrescribing data.

The analysis demonstrated:

- sustained antidepressant prescribing demand
- measurable variation across NHS organisations
- important prescribing cost relationships
- strong suitability of OpenPrescribing data for healthcare analytics workflows

Although anxiolytic-related fields contained partial sparsity within the supplied dataset, the overall analytical pipeline remained effective for trend analysis and prescribing intelligence generation.

The outputs generated through this workflow provide a strong foundation for future:

- forecasting analysis
- NHS prescribing analytics
- regional healthcare studies
- medication utilisation modelling
- public health reporting